Unit – II
Sets, Relations, Orderings, and Functions

1. Show that $(A \cap B) \cup C = A \cap (B \cup C)$ iff $C \subseteq A$

2. Let X = { 1,2,3,4} and R = {(x,y) / x > y}. Draw the graph of R and also give its matrix.

3. Let A be a given finite set and P(A) its power set. Draw Hasse diagram of $(P(A) , \subseteq)$ for A = { *a,b,c,d*}.

4. Prove that for any three sets A,B,and C,
(i)$A\times (B\cup C) = (A\times B) \cup (A\times C)$
(ii)$A\times (B\cap C) = (A\times B) \cap (A\times C)$

5. Let X = { 1,2,...,7} and R = { (x,y) | x-y is divisible by 3}. Show that R is an equivalence relation. Draw the graph of R.

6. Given S = {1,2,…,10} and a relation R on S where R = { (x,y) | x +y =10}. What are the properties of the relation R ?

7. If R is a partial ordering relation on a set X and $A \subseteq X$, show that $R \cap (A\times A)$ is a partial ordering relation on A.

8. Let X = {2,3,6,12,24,36} and the relation ≤ be such that $x \leq y$ if x divides y. Draw the Hasse diagram of $(X, \leq)$.

9. Show that the relation R = {(a, b) | $a \equiv b \pmod m$}, where m be an integer with m > 1 is an equivalence relation on the set of integers.

10. Let R denote a relation on the set of ordered pairs of positive integers such that (x,y) R (u,v) iff xv=yu. Show that R is an equivalence relation.

11. Show that there are only five distinct Hasse diagrams for partially ordered sets that contain three elements.

12. Define the following in a Poset
    (i) least member
    (ii) greatest member
    (iii) minimal member
    (iv) maximal member

13. Discuss the concept of functions, including definitions, domain, range, and types of functions.

14. Let X = {1, 2, 3} and f, g, h and s are functions from X to X defined by f = {(1, 2), (2, 3), (3, 1)}, g = {(1, 2), (2, 1), (3, 3)}, h = {(1, 1), (2, 2), (3, 1)} s = {(1, 1), (2, 2), (3, 3)}. Then find fog, fos, fohog and sog.

15. Explain the concept of inverse functions in detail, including the conditions for a function to be invertible.

16. Let $f : R \rightarrow R$ and $g : R \rightarrow R$, where R is the set of real numbers. Then find fog and gof where $f(x) = x^2 - 2$ and $g(x) = x + 4$. State whether these functions are injective, surjective, and bijective.

17. Define one-one and onto functions. Show that the functions $f(x) = x^2$ and $g(x) = x^{1/2}$, for $x \in [0, \infty)$, are inverse of one another.

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